END-TO-END DATA ANALYTICS PROJECT

# Customer Shopping Behavior Analysis

Python · PostgreSQL · Power BI | 8,000 Transactions · ₹217M Revenue · March 2026

OVERVIEW

# Business Problem Statement

A leading retail company wants to better understand customer shopping behavior to improve **sales, satisfaction, and loyalty.**

> "How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

## Key Challenges

### Shifting Patterns

Purchase behavior shifting across demographics & channels

### Discount Dilemma

Are discounts helping revenue or hurting margins?

### Retention Drivers

What drives repeat purchases & long-term retention?

### Customer Value

Which customers are most valuable to the business?

METHODOLOGY

# Dataset & Methodology

## Dataset Overview

- 8,000 records × 19 columns (raw) → 21 columns (cleaned)
- 4 product categories: Clothing, Electronics, Accessories, Footwear
- 8 unique locations | Date range: 2024–2025
- Purchase Amount: ₹1,245 – ₹1,24,251

## Tools Used

- **Python:** Data cleaning, EDA, feature engineering
- **PostgreSQL:** 16 business queries + RFM + Cohort + CLV
- **Power BI:** 3-page interactive dashboard with DAX

## Data Cleaning Highlights

### Review Rating

120 nulls → median imputed per category

### Shipping Type

100 nulls → mode imputed per category

### Size

5,479 nulls (68.5%) → handled appropriately

### New Features

Engineered: `age_group` & `purchase_frequency_days`

KPIS

# Key Performance Metrics

**₹217M**
Total Revenue
Across 8,000 customers

**₹27K**
Avg Order Value
₹27,149 per transaction

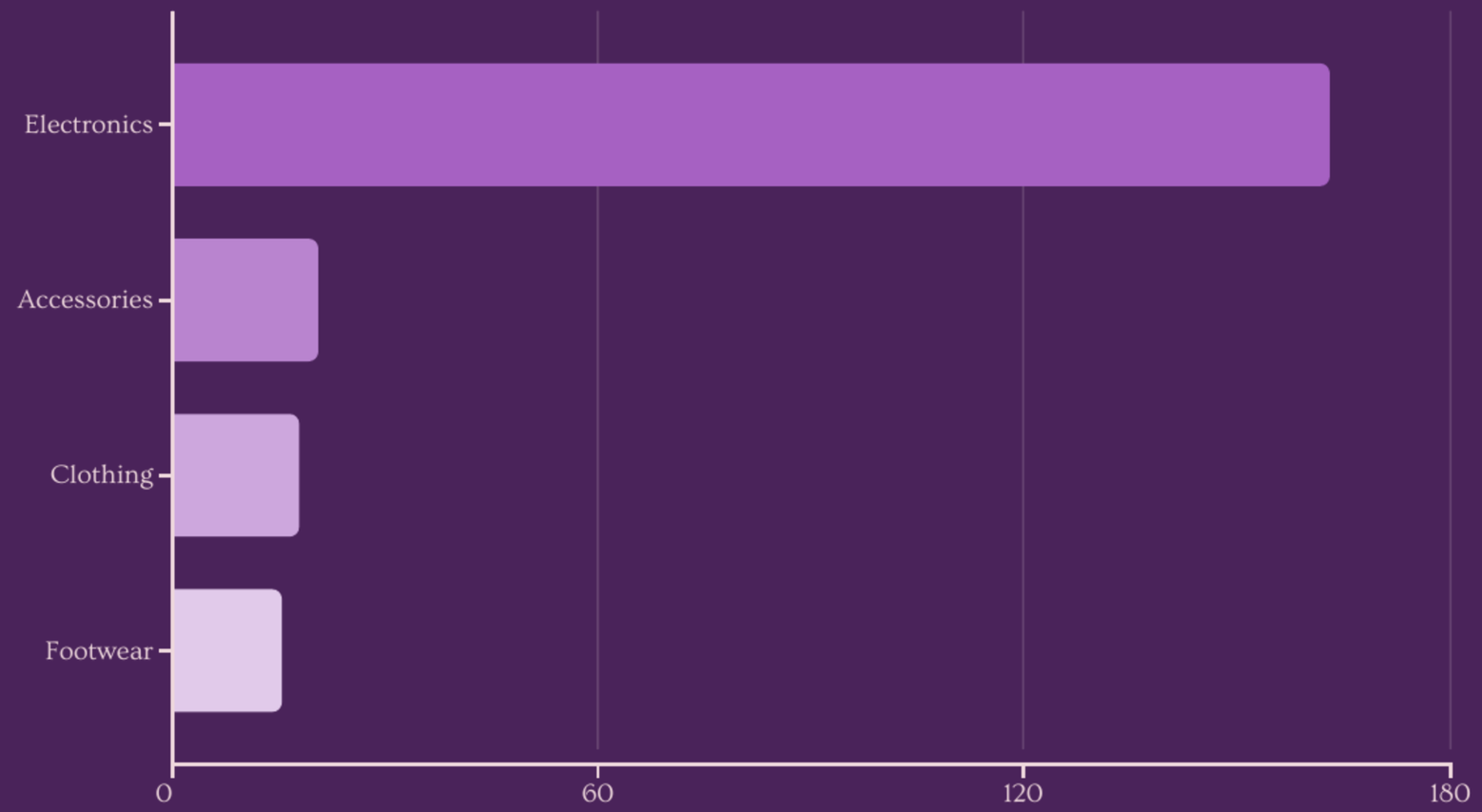
**42%**
Repeat Purchase Rate
Returning customer share

**40.2%**
Discount Usage Rate
Potential margin risk

**3.97**
Avg Review Rating
Out of 5.0

**68**
Avg Purchase Interval
Days between purchases

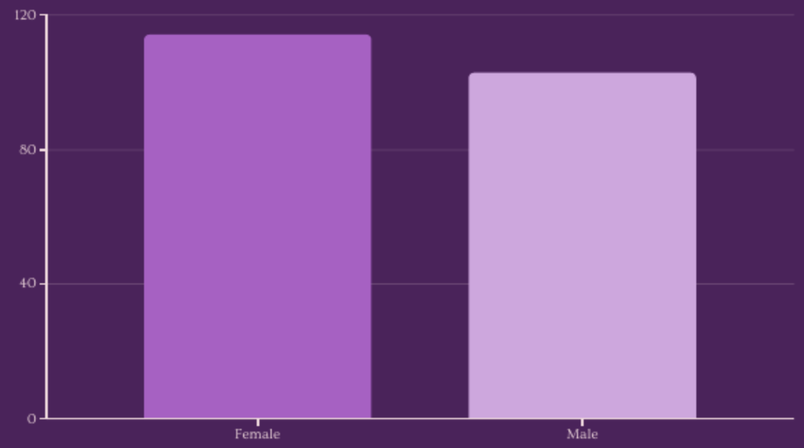
Revenue by Category



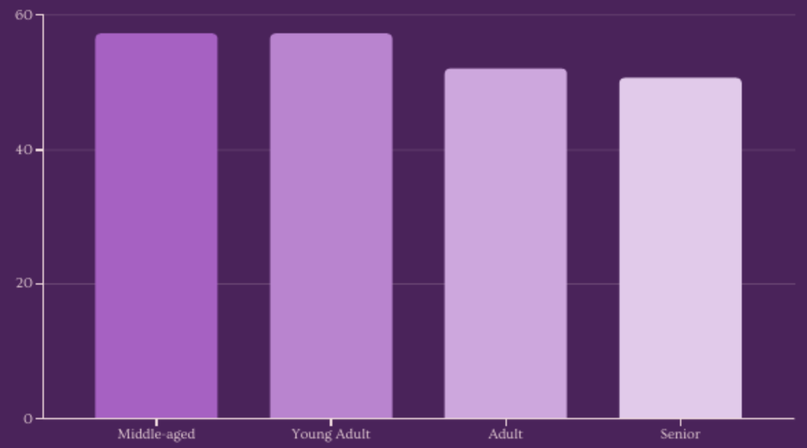
EDA RESULTS

# Customer Behavior Insights

## Revenue by Gender




## Revenue by Age Group




Female customers (4,180) contribute **₹114.2M** vs Male (3,820) at **₹102.9M** – defying common assumptions.

### Only 41 Highly Loyal

Customers with 20+ purchases out of 8,000 total

### Winter Leads Spend

Highest avg ₹27,730 – all seasons within 5% range

### Top Payment Method

Credit Card 40.7% | Debit 24.8% | PayPal 14.7%

### Recency Effect

0–30 day recency customers show highest spend consistently

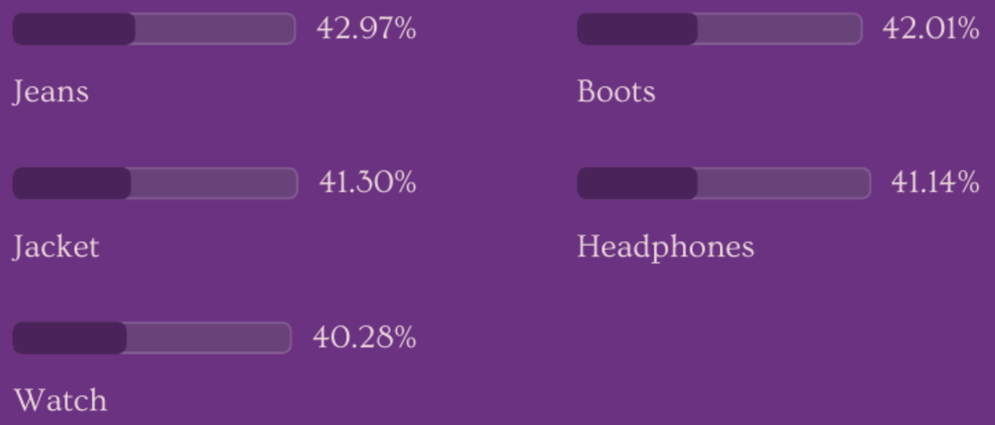
REVENUE ANALYSIS

# Discount & Subscription Analysis

## Discount Impact on Revenue

40.2% discount rate is **hurting margins** – non-discounted revenue exceeds discounted by ₹46.4M.

## Top Discount-Dependent Products




## Subscription & Promo Code Findings

### Subscription Program

Subscribers (2,763): Avg ₹27,319 | Non-subscribers (5,237): Avg ₹27,058

Only **₹261 difference** – program not differentiating value

### Promo Code Impact

With promo: 10.04 avg repeat purchases | Without: 10.01

Promo codes do **NOT** build long-term loyalty – proven by data

SEGMENTATION

# RFM Segmentation Model

Customers scored on **Recency** · **Frequency** · **Monetary value** using NTILE(4) quartile method in PostgreSQL.

### Champions (Score 10–12)

Buy recently, often & spend most

→ VIP rewards & early access

### Loyal Customers (Score 7–9)

Regular buyers with good spend

→ Upsell & cross-sell

### Potential Loyalists (Score 4–6)

Moderate engagement, growing

→ Targeted promotions

### At Risk (Score ⩽ 3)

Haven't purchased recently

→ Win-back campaigns

## Advanced Analysis Performed

### Cohort Retention

Monthly retention tracked from first purchase using `DATE_TRUNC`

### Customer Lifetime Value

Top 20 customers ranked by cumulative lifetime spend for VIP targeting

SQL HIGHLIGHTS

# Key SQL Analysis Results

### Q1 – Revenue by Gender

Female **₹114.2M** vs Male **₹102.9M**

### Q3 – Top Rated Products

Headphones & Boots (4.00), Backpack (3.99)

### Q6 – Revenue by Age Group

Middle-aged **₹57.3M** leads all segments

### Q9 – Seasonal Behavior

Winter highest **₹27,730** – consistent across seasons

### Q11 – Loyalty Segments

Moderately Loyal 4,270 | Low Engagement 3,689 | **Highly Loyal only 41**

### Q12 – Promo Code Impact

Near-zero loyalty lift: 10.04 vs 10.01 avg repeat purchases

### Q14 – Payment Maturity

Cash users oldest avg age 44.07 | Google Pay youngest 42.85

### CLV Top 20

Highest lifetime value customers identified for **VIP targeting**

STRATEGY

# Business Recommendations

### VIP for Champions

Launch VIP program to protect top revenue

### Win-back At-Risk

Recover customers using recency-focused outreach

### Convert Loyalists

Target 41% base for fastest growth

### Redesign Subscription

Close ₹261 value gap to improve uptake

### Reduce Discount Reliance

Limit promos on Jeans, Boots, Jackets

Data-driven priorities to protect Champions, grow Potential Loyalists, and fix margin-eroding discount and subscription strategies.

CONCLUSION

# Key Takeaways

### → Champions Drive Revenue

39% of customers generate **82% of ₹217M** – protecting this segment is priority #1

### → Discounts Are Margin-Negative

40.2% discount rate yields ₹85.4M vs ₹131.8M non-discounted – structural change needed

### → Promos Attract, Not Retain

Promo codes attract one-time buyers, not loyal customers – proven by near-zero lift data

### → Subscription Needs Redesign

Only ₹261 avg spend difference – program must deliver stronger differentiated value

### → Recency Is the #1 Predictor

0–30 day recency customers consistently spend the most – re-engagement is key

### → Loyalty Program Urgently Needed

Only 41 Highly Loyal customers out of 8,000 – massive opportunity gap remains

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**Skills Demonstrated:** Python EDA · Advanced SQL (CTEs, Window Functions) · RFM + Cohort + CLV Modeling · Power BI Dashboard · Business Insight Translation · End-to-End Project Delivery